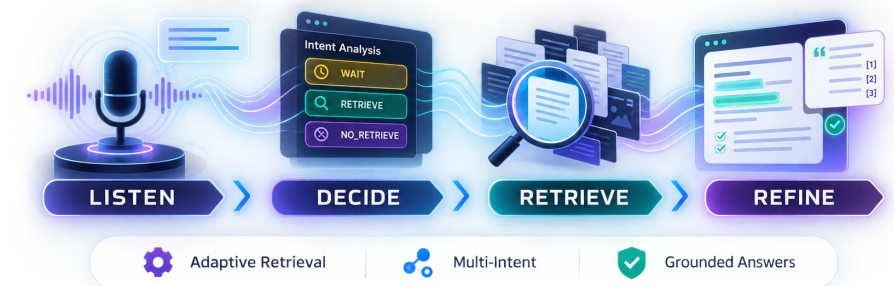
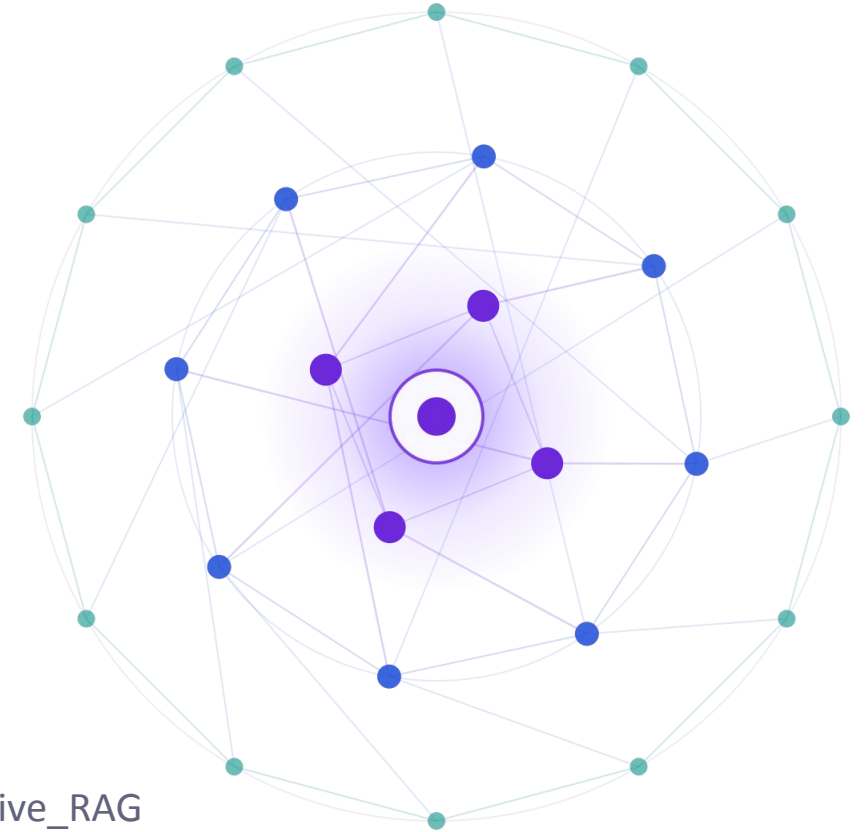


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - Theme 4 — Streaming Live RAG
- Team Name - GenX
- College Name - Thapar Institute of Engineering & Technology
- Member Name & Email 1- Prabhasees Singh & psingh12_be25@thapar.edu
- Member Name & Email 2- Krishnansh Goswami & kgoswami_be22@thapar.edu
- Submission Github link- https://github.com/prabhaseessingh/GenX_Streaming_Live_RAG



Theme-4: Streaming Live RAG

The core challenge:

Information arrives **incrementally**, so RAG must decide **when to retrieve, what to retrieve, and when to refine** as intent evolves.

GenX implements:

- **Incremental Input** — process transcript chunks as they arrive
- **Adaptive Retrieval** — **WAIT / RETRIEVE / NO_RETRIEVE**
- **Multi-Intent Decomposition** — split compound requests into subqueries
- **Parallel Retrieval** — search multiple intents concurrently
- **Session Refinement** — use **delta retrieval** for new constraints
- **Grounded Safety** — validate answers against retrieved evidence
- **Observable RAG** — track decisions, retrieval, validation and refinement

Current implementation:

Uploaded Audio → Faster-Whisper → Timestamped ASR → Incremental RAG

Not yet live microphone streaming or token-level LLM streaming.

Existing Solutions & Gaps

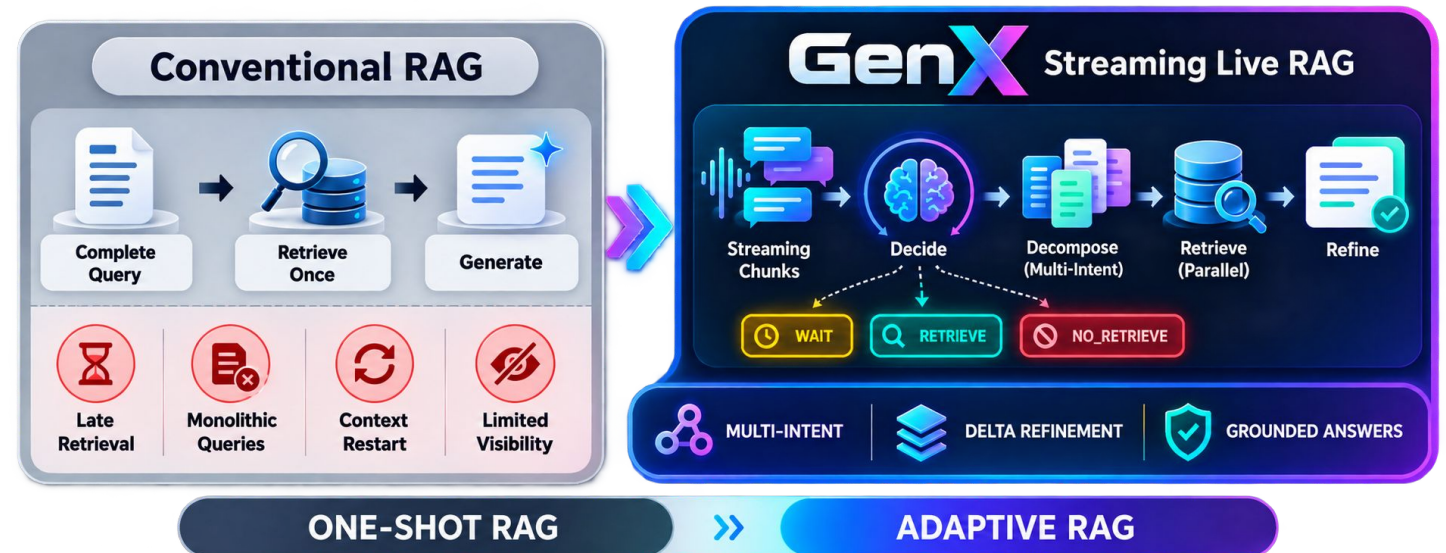
Existing Gap

Conventional RAG typically waits for a complete query and treats retrieval as a one-shot operation.

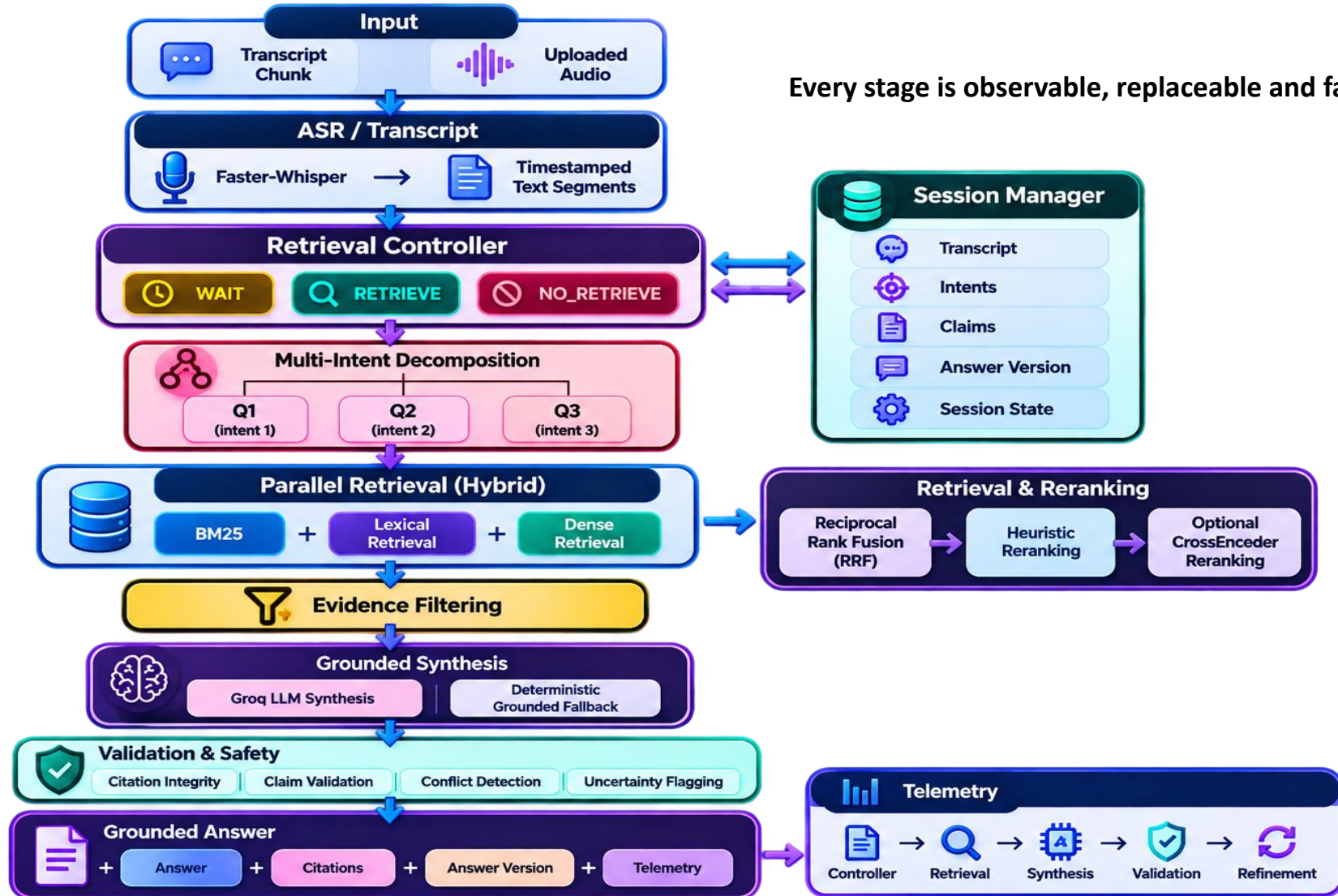
GenX Approach

- Conditional retrieval
- Parallel multi-intent search
- State-preserving refinement
- Evidence-first generation
- Grounding & conflict checks
- Observable decision trail

From one-shot RAG to adaptive RAG.



Our Solutions & Architecture Diagram



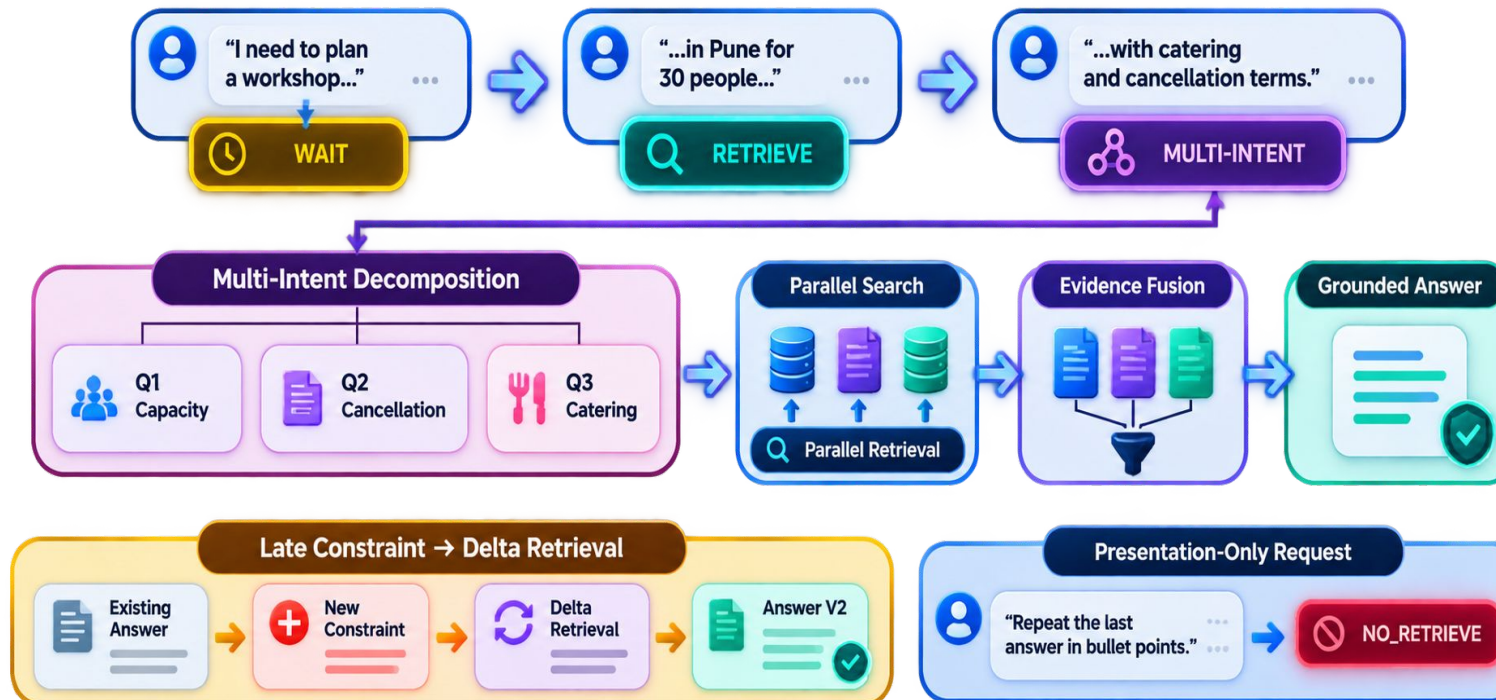
Demo & Product Walkthrough

➤ Incremental Retrieval

Incomplete input → **WAIT** → stable intent → **RETRIEVE**

➤ Multi-Intent

Compound query → **multiple subqueries** → **parallel retrieval**



➤ No Retrieval

"Repeat in bullet points" → **NO_RETRIEVE**

➤ Refinement

New constraint → **delta retrieval** → **selective update**

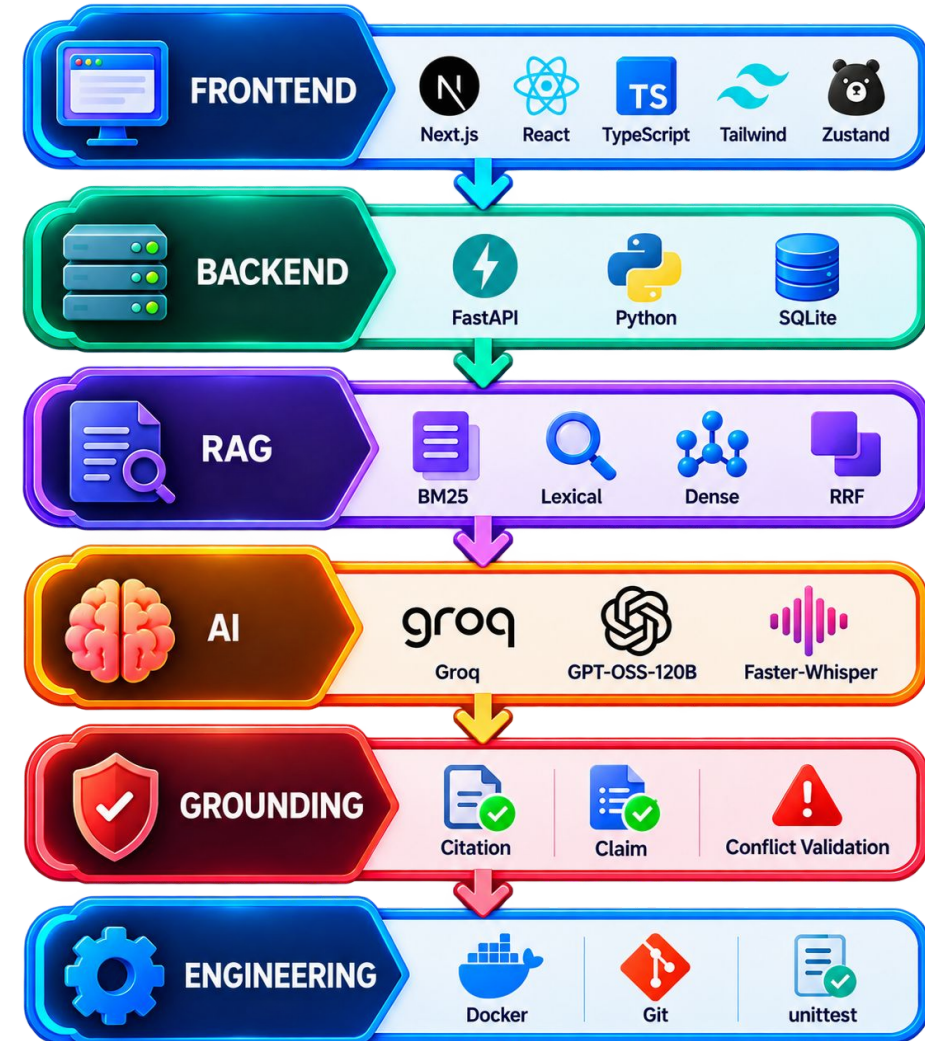
➤ Unsupported Query

Insufficient evidence → **grounded fallback**

Tools and tech stack used

Built for Adaptive, Grounded RAG

- **Interface:** Next.js + React for the interactive RAG workspace.
- **Backend:** FastAPI orchestrates sessions, retrieval and APIs.
- **Retrieval:** BM25 + lexical + dense search fused with RRF.
- **AI:** Faster-Whisper for ASR; Groq / GPT-OSS-120B for synthesis.
- **Grounding:** Citation, claim-support and conflict validation.
- **Engineering:** Docker + Git + **34 automated tests**.



Impact & Use case

❖ Current Evaluation

100% Top-5 Hit Rate

62.5% Precision@5

72.2% Recall@5

61.7% MRR

Local supplied corpus.

Use Cases



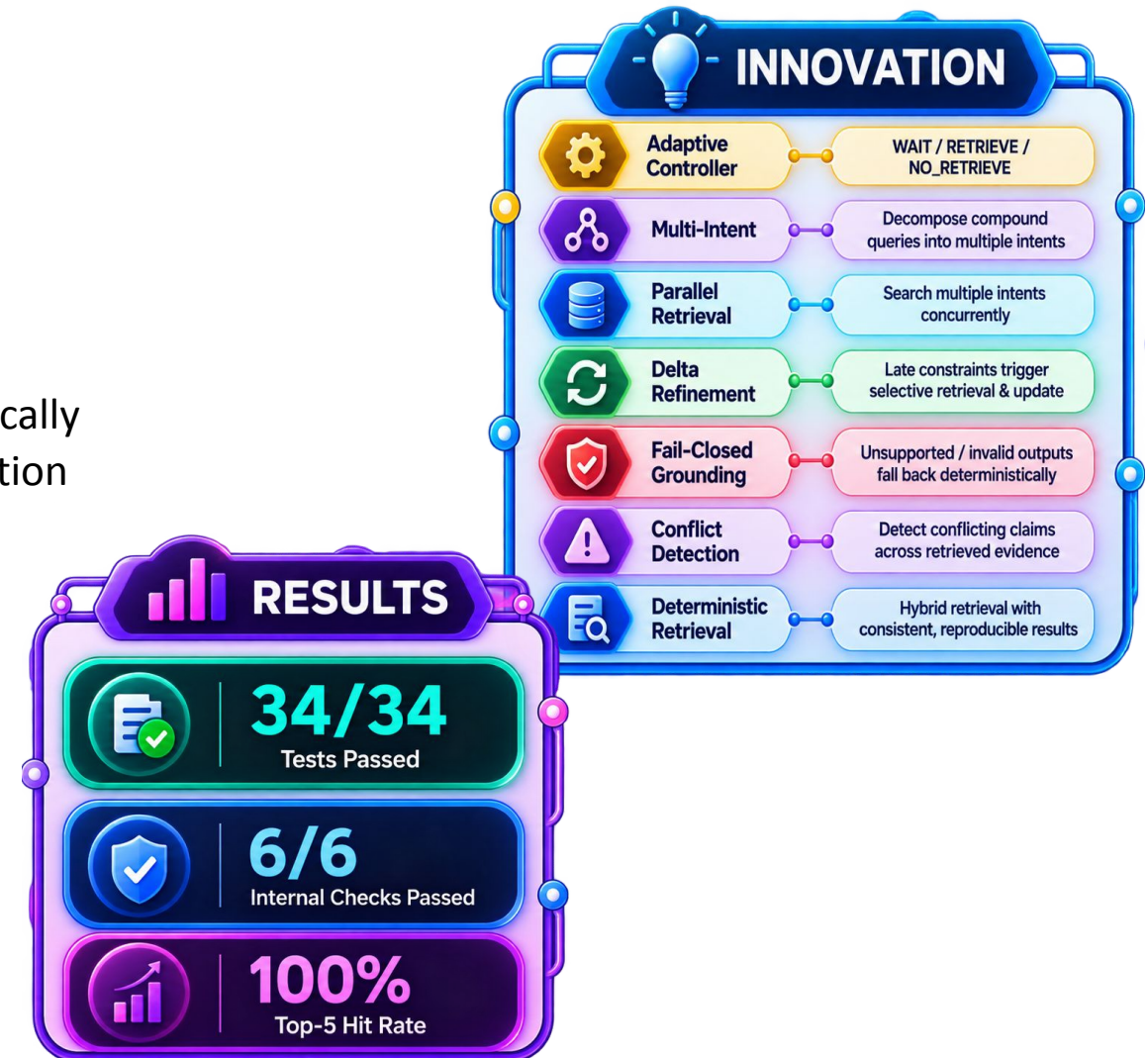
Innovation highlights, results and limitations

Key Innovations

- ❑ **Adaptive:** WAIT / RETRIEVE / NO_RETRIEVE
- ❑ **Parallel:** multiple intents searched concurrently
- ❑ **Stateful:** late constraints trigger delta retrieval
- ❑ **Grounded:** evidence validated before final response
- ❑ **Safe:** unsupported / invalid outputs fall back deterministically
- ❑ **Observable:** controller → retrieval → synthesis → validation

Current Limitations

- Official held-out corpus not yet benchmarked
- Live microphone streaming not implemented
- Token-level LLM streaming not implemented
- CrossEncoder / NLI are optional
- Formal ablations remain



What's next

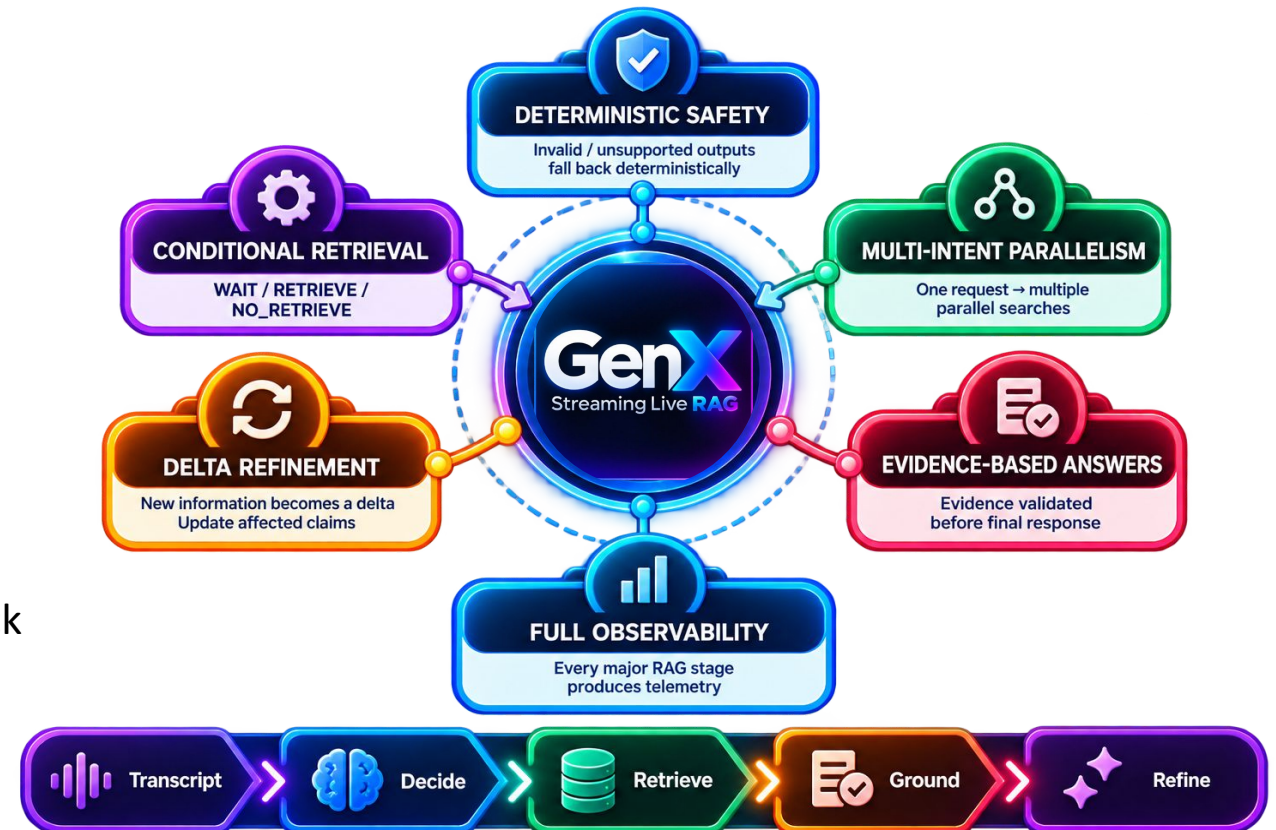
- Validate: *Official corpus • latency • grounding metrics*
- Benchmark: *Retrieval • controller • reranker • validator ablations*
- Strengthen: *Intent stability • semantic reranking • claim entailment • delta detection*
- True Live Streaming: *Microphone input • streaming ASR • stream-aware synthesis*
- Production: *Authentication • rate limiting • CI gates • scalable sessions*



Brownie points slide(differentiation)

Why GenX?

- ❑ **01 — Retrieval is conditional**
WAIT / RETRIEVE / NO_RETRIEVE
- ❑ **02 — Compound intent is first-class**
One request → multiple parallel searches
- ❑ **03 — New information becomes a delta**
Update affected claims, preserve session state
- ❑ **04 — Evidence controls generation**
Invalid / unsupported output → grounded fallback
- ❑ **05 — The pipeline is observable**
Every major RAG stage produces telemetry



GenX treats RAG as an adaptive state machine — not a single search call behind an LLM.

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N)- **YES**
- README with reproducible setup instructions (Y/N)- **YES**
- Demo video, max 5 minutes (YouTube or Drive link)- **YES**
- Presentation file (PPT or PDF) (Y/N)- **YES**

Thank you :)

GenX Streaming Live RAG

Team GenX

Thapar Institute of Engineering & Technology

GitHub:

https://github.com/prabhaseessingh/GenX_Streaming_Live_RAG

Demo:

<https://drive.google.com/file/d/1O0S11cT4RK201Q9DA3Om2AzDHGLNSCC7/view?usp=sharing>



Don't wait for the question to end. Decide when the evidence is ready.